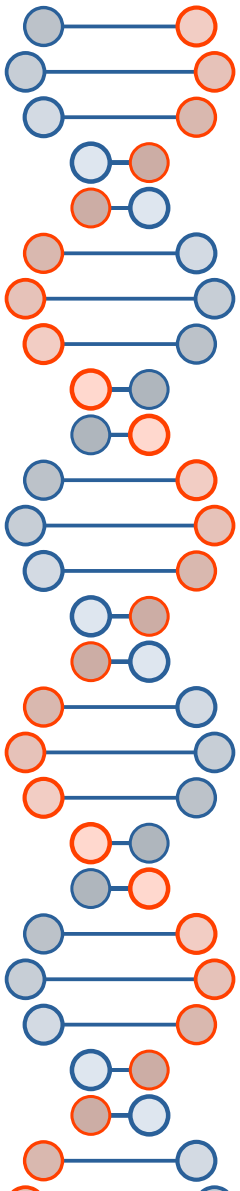


PCR

Immunology – Lab 1

Leonard Mada



PCR: Introduction

Polymerase Chain Reaction:

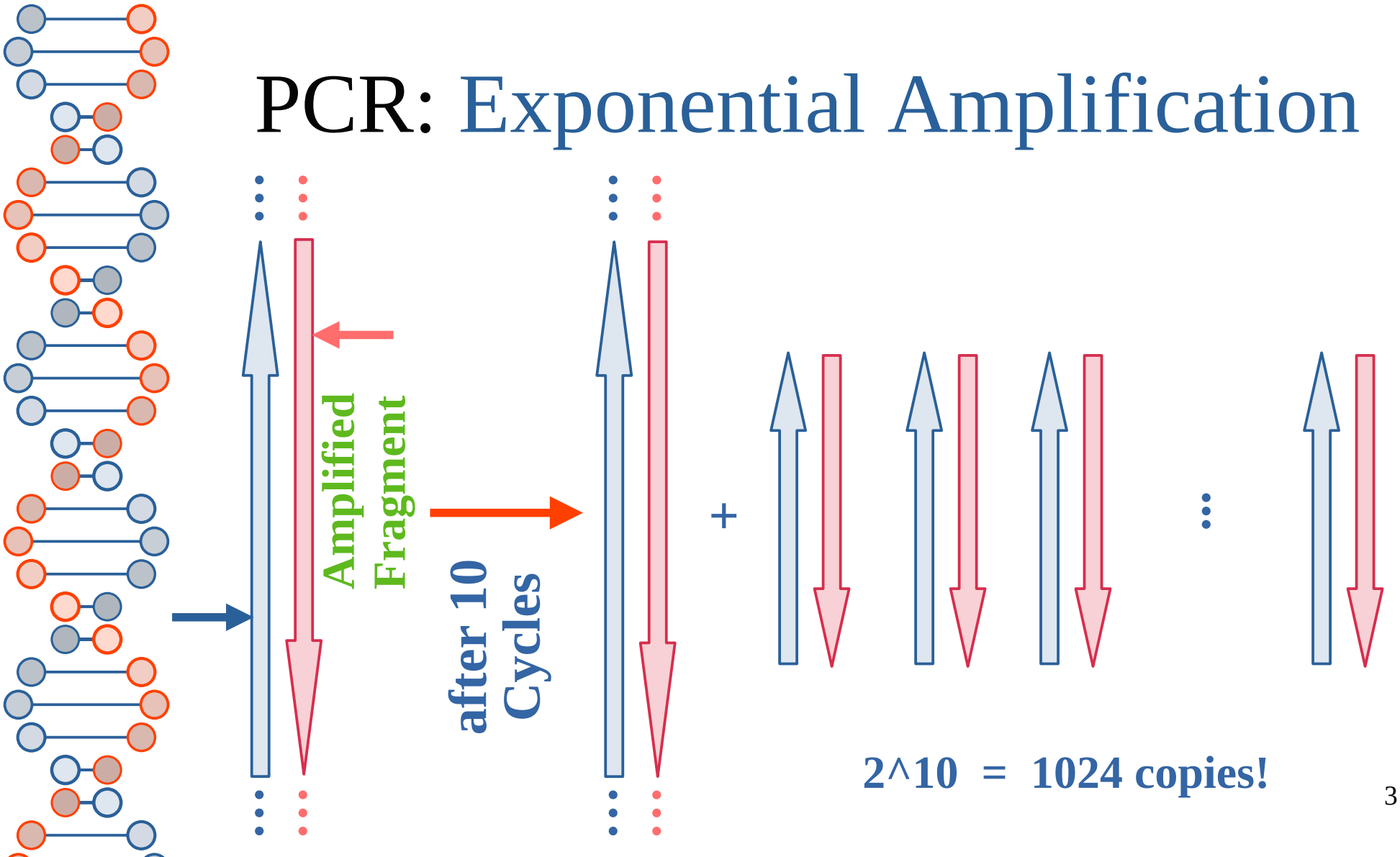
- **Amplification** of a **specific** DNA region using **polymerase**;

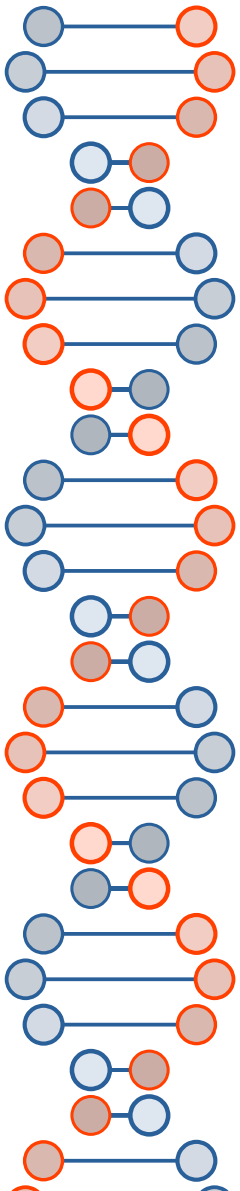
History:

- Developed in 1983 – 1985; Nobel for Chemistry (1993);
- Mullis & Fallona: Specific Enzymatic Amplification of DNA in vitro: The Polymerase Chain Reaction. (1986)

Veritasium: https://www.youtube.com/watch?v=zaXKQ70q4KQ

PCR: Exponential Amplification





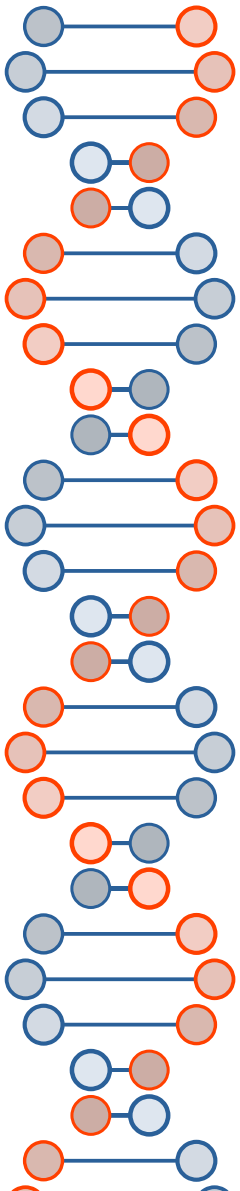
PCR: Characteristics

Advantages:

- **Rapid**: 2 hours;
- **Simple**: very simple lab procedures;
- **Apparatus**: only 1 specialized device (thermocycler);
- **Reagents**: + 2 primers (oligonucleotides) – **must be synthesized**;

Applications:

- **Selective amplification** of DNA: eg. gene fragment;
- **Sequence**: **known** or **unknown**; any organism;



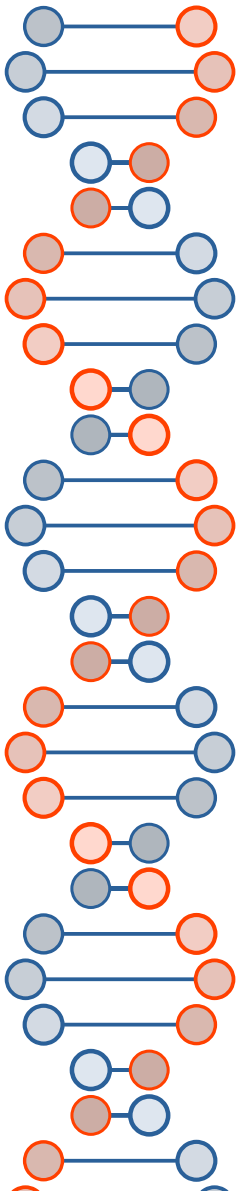
PCR: Method

Steps:

- **Step 1**: **DNA** Extraction (1 hour – hours)
- **Step 2**: Amplification (**PCR**); (2 h)
- **Step 3**: Visualization of amplified fragment; (1-2 h)

Variants:

- **RNA** => rtPCR (reverse-transcription PCR);
- **Qantitative**: qPCR / real-time PCR;



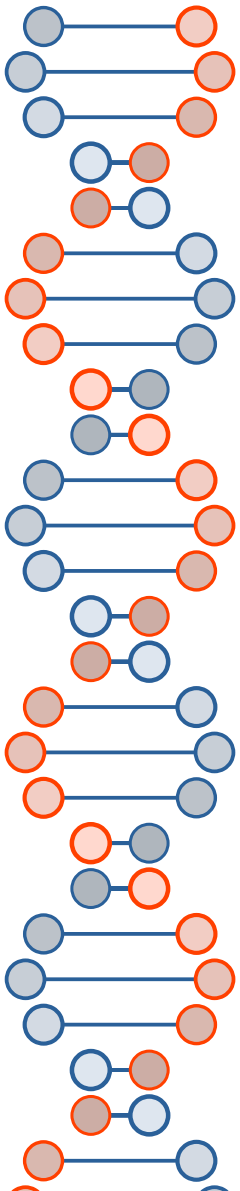
PCR: Method

Step 1: DNA Extraction

- **Source**: blood, tissues, cell culture, bacteria, sputum, other;
- **Quantity**: a few **DNA** molecules; (1 nano-g DNA)
- **Genome**: DNA can be fragmented;
- **Extraction**: Commercial kits (e.g. cell lysis, etc);

Variant: rtPCR (reverse transcription PCR)

- **RNA** => **revers-transcriptase** => **cDNA** => **dsDNA**=> PCR;
- Source: **viral RNA** or **mRNA** (transcriptomics);



PCR: Method

Step 2: Amplification

- **Exponential** growth of amplified **DNA** fragment;
- 1 molecule => 2^30 molecules ≈ **1 billion**;
- **Warning: cannot** quantify **initial quantity**!

Variants:

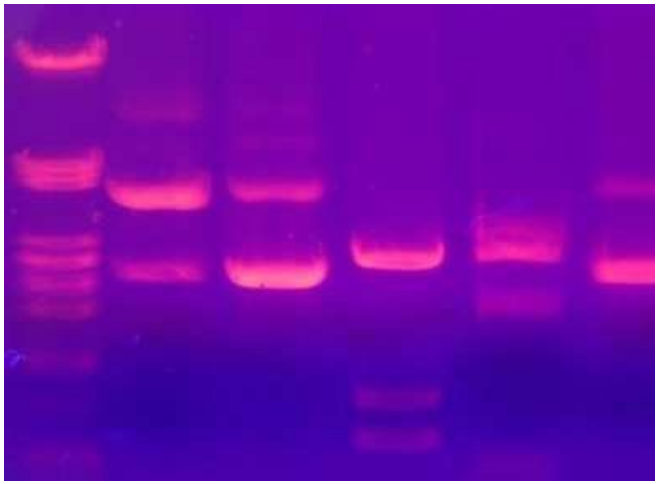
- **rtPCR**: **RNA** => **revers-transcriptase** => **cDNA**;
- **qPCR**: **quantitative** PCR = real-time PCR;

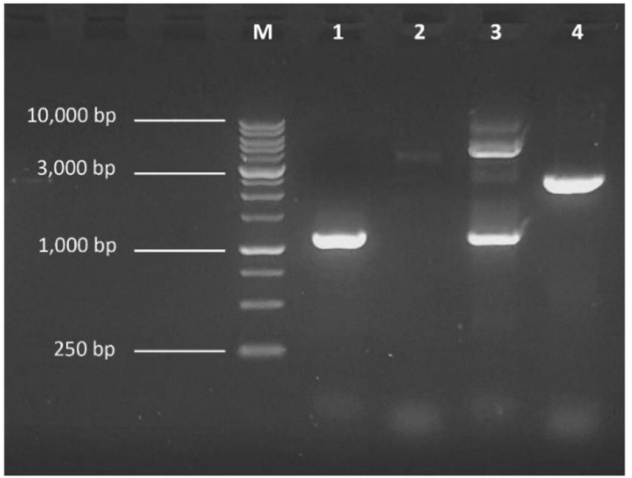
PCR: Method

Step 3: Visualization

- **Gel Electrophoresis**: separation of DNA fragments; (1-2 h)
- **Agarose**: 0.5% - 1% - 1.5%; (classic)
- **UV**: *ethidium bromide* (carcinogen)!

DNA Ladder

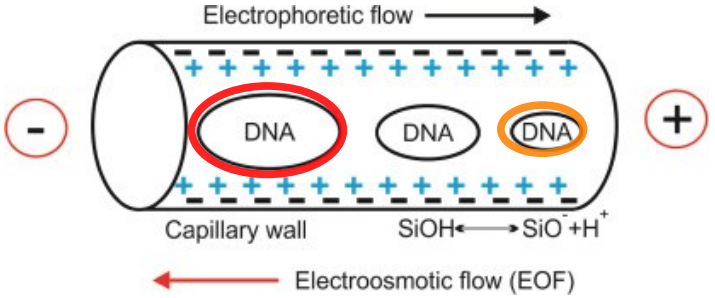


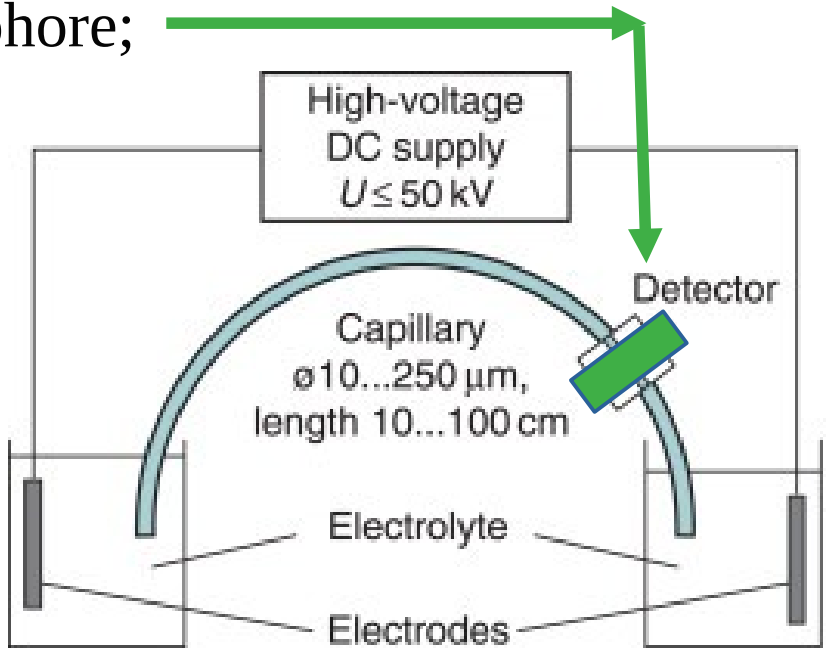


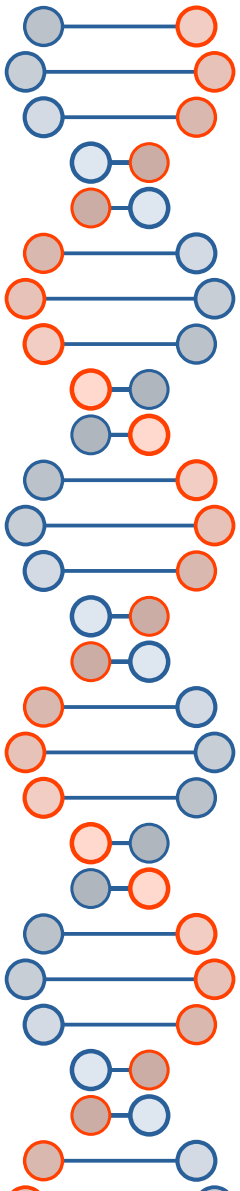
PCR: Method

Step 3: Capillary Electrophoresis

- **Primers** marked w. fluorophore;
- **Fast**: 10,000 V and more;
- **Minimal** DNA quantity;







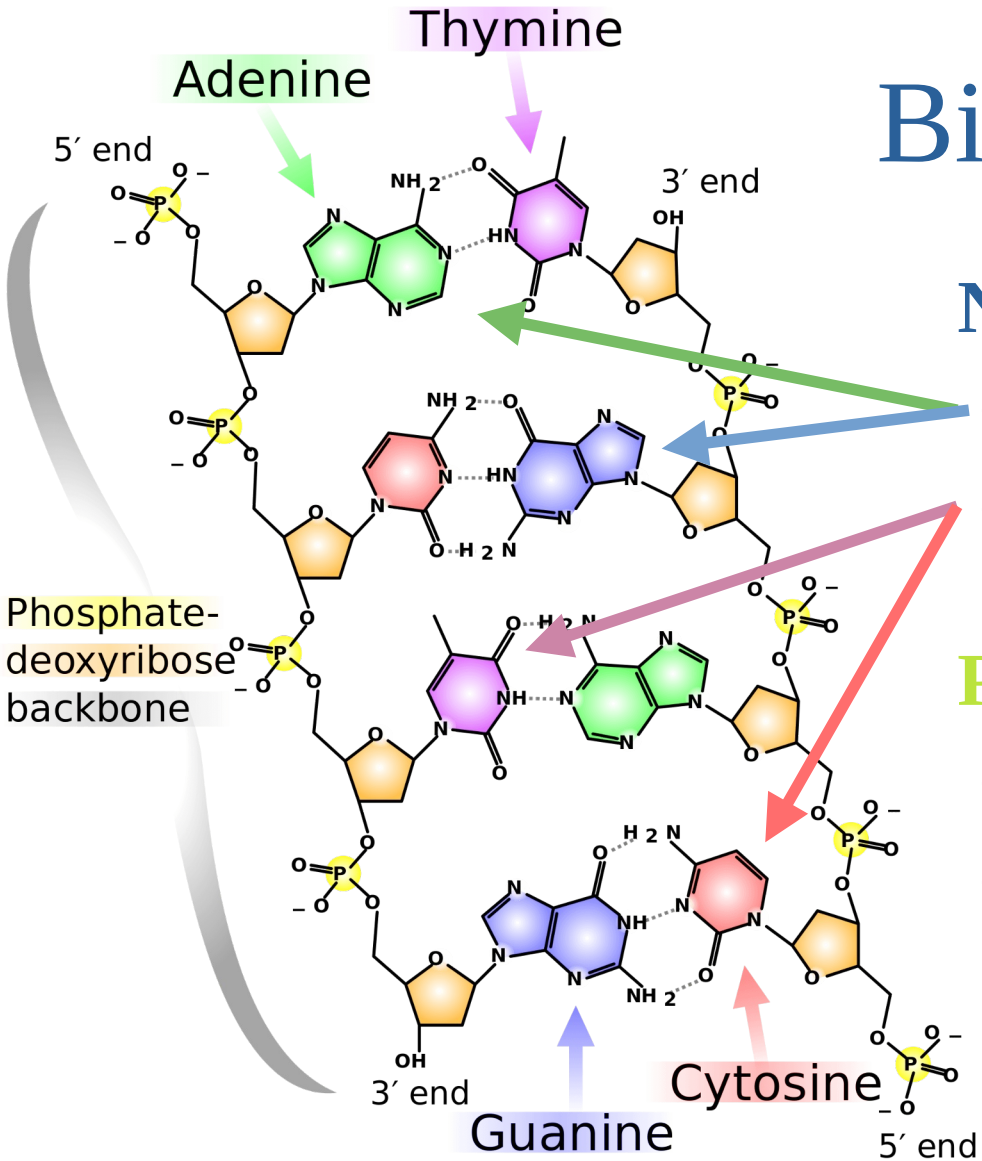
PCR: Method

Step 3: Visualization

Other methods:

- Hybridization w. **labeled probes** (are similar to primers);
  - Specific oligonucleotides: high **specificity**!
- Radioactive labeling: auto-radiography; (rare)
  - in general not necessary: sufficient amplification by PCR!
  - PCR may be used to synthesize RX probes;

Biochemistry



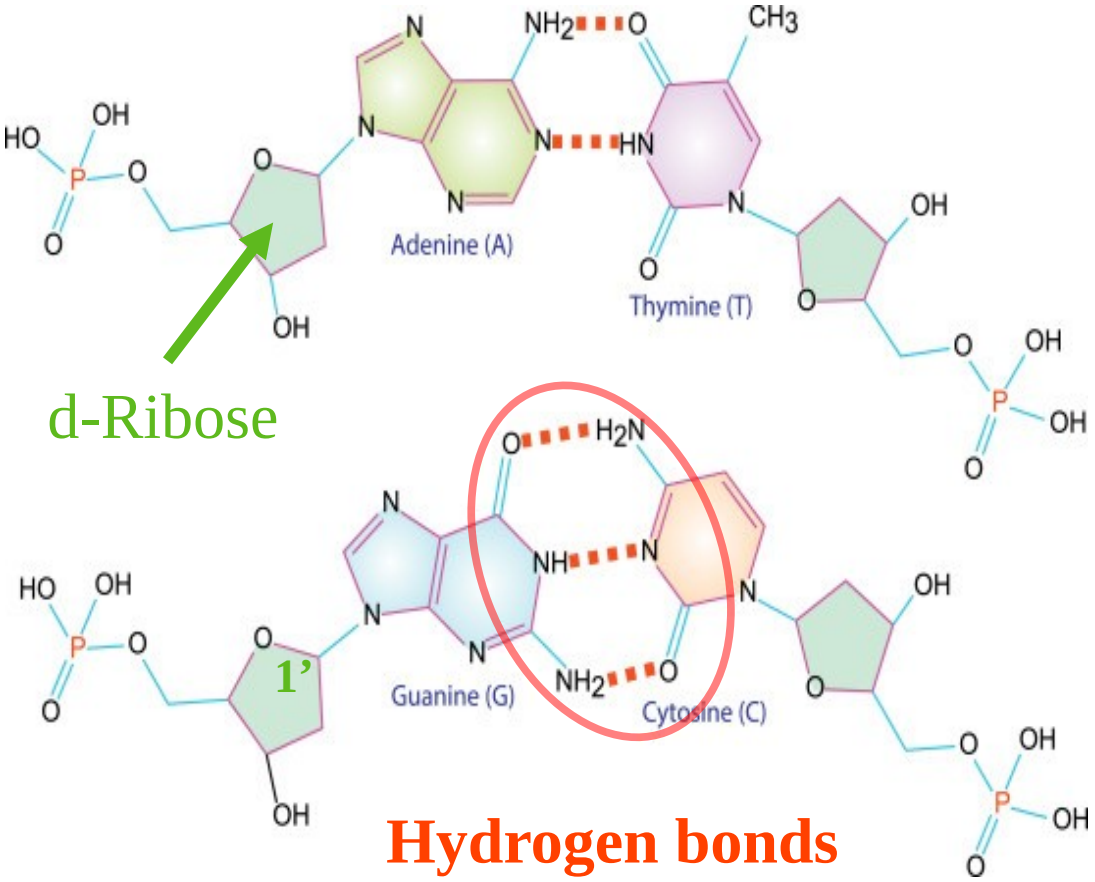
Nucleotides:

- **Purines**: G, A
- **Pyrimidines**: C, T

Pairs: H-bonds

- **G ≡ C**
- **A = T**

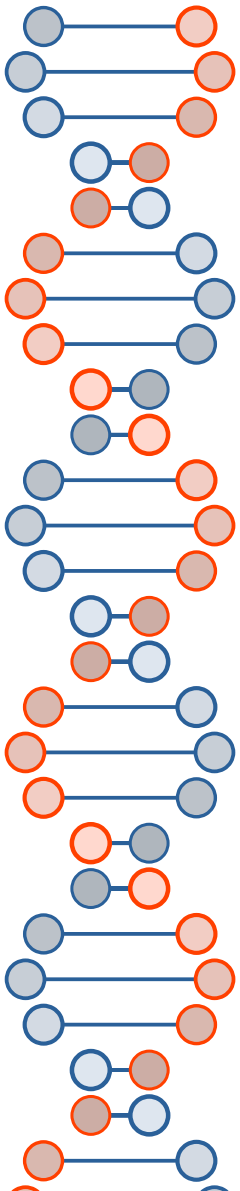
PCR: Base Pairs



Nucleotides:

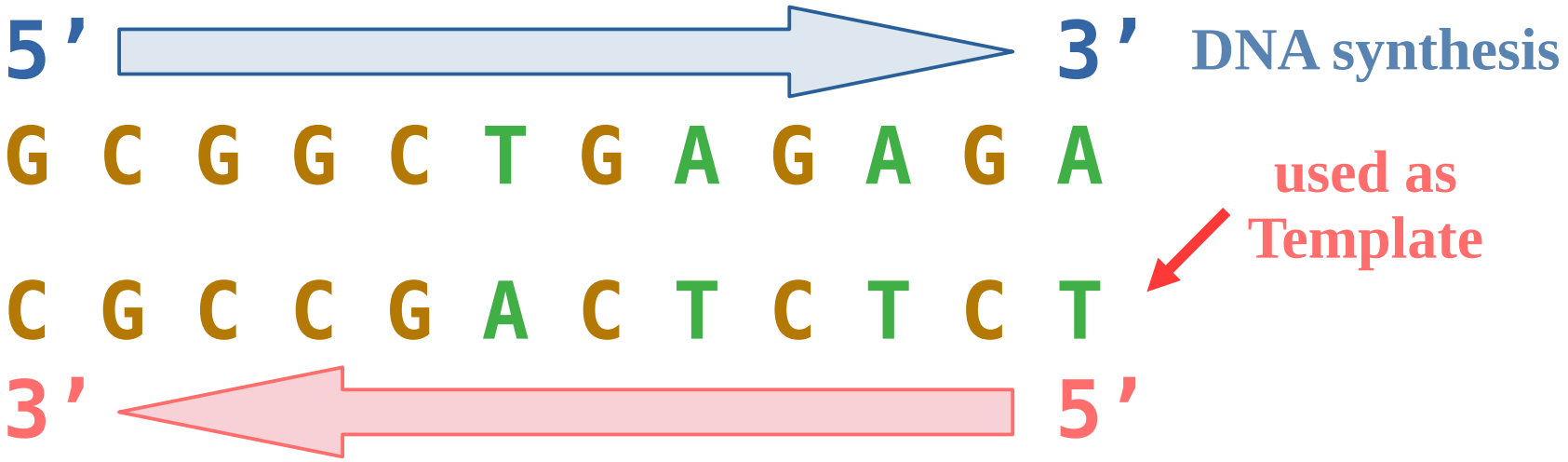
A ≡≡≡ T

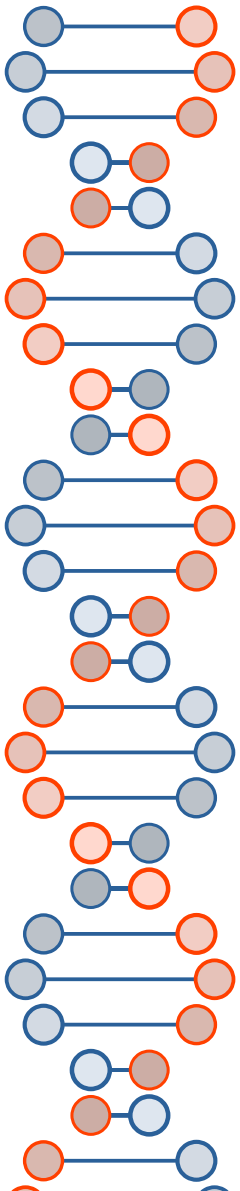
G ≡≡≡ C



PCR: DNA Structure

- 2 Complementary strands: G vs C, A vs T;
- **Orientation:** 5’ => 3’





PCR: Technique

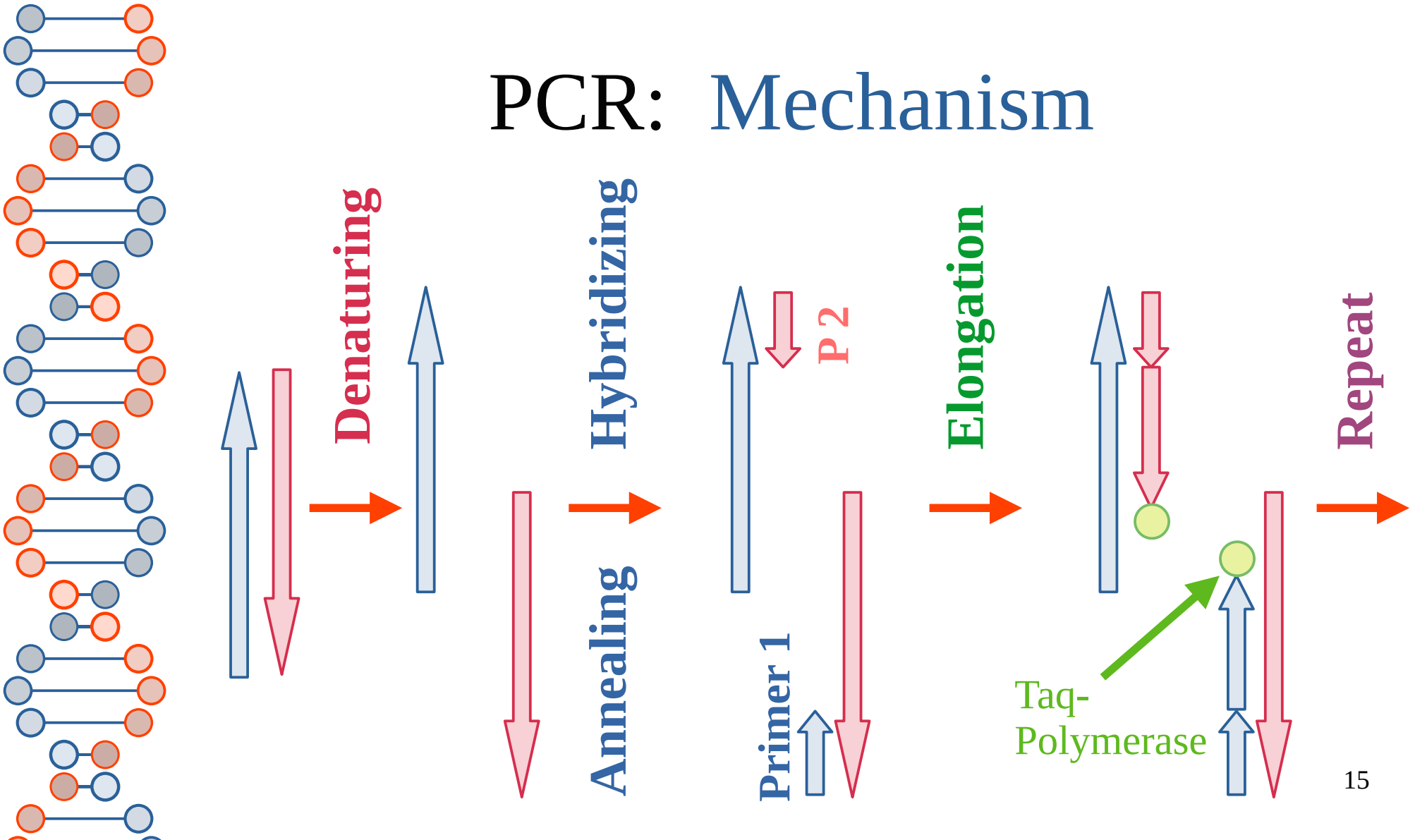
PCR: 1 Cycle

- **Step 1**: **Denature** DNA; 92 – 95º C;
- **Step 2**: **Hybridize** primers; 40 – 65º C;
- **Step 3**: **Extend** primers; 68 – 72º C;
- Repeat cycle: 30 cycles;

Extra Steps:

- Denaturation: 2 minutes (before Cycle 1);
- Extension: 2 minute (after last Cycle);

PCR: Mechanism



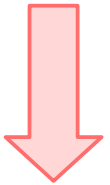
PCR: Technique

Step 1: DNA Denaturing

- **Temperature**: 92 – 95° C;
- **Duration**: ~ 30 seconds;
- Before first cycle: 1 – 2 minutes;

...G C G G C T G A G A G A...

...C G C C G A C T C T C T...

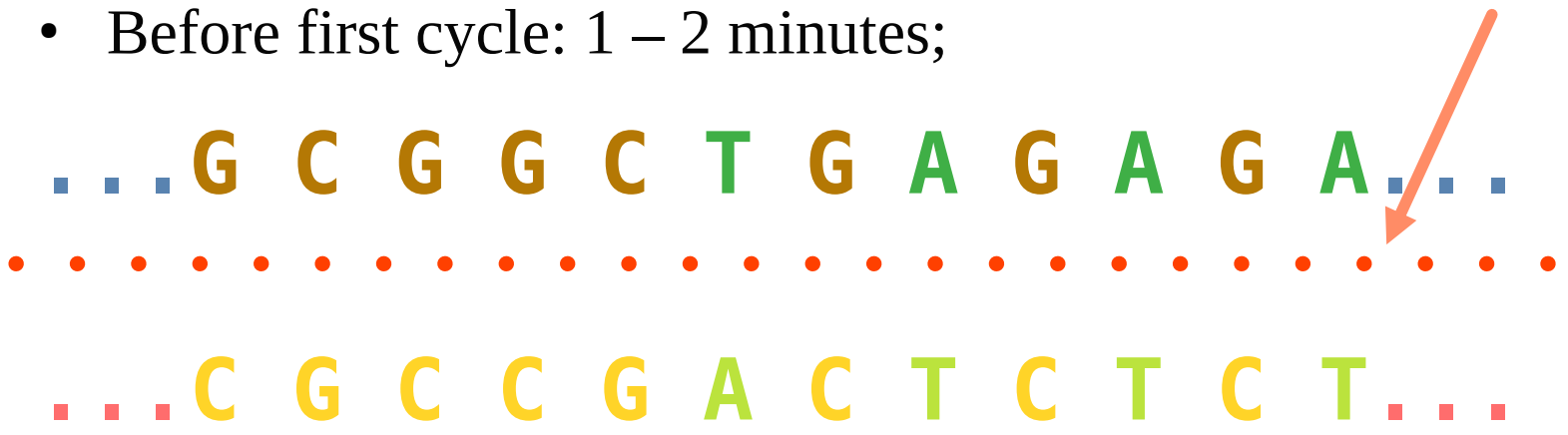


PCR: Technique

Step 1: DNA Denaturing

- **Temperature**: 92 – 95º C;
- **Duration**: ~ 30 seconds;
- Before first cycle: 1 – 2 minutes;

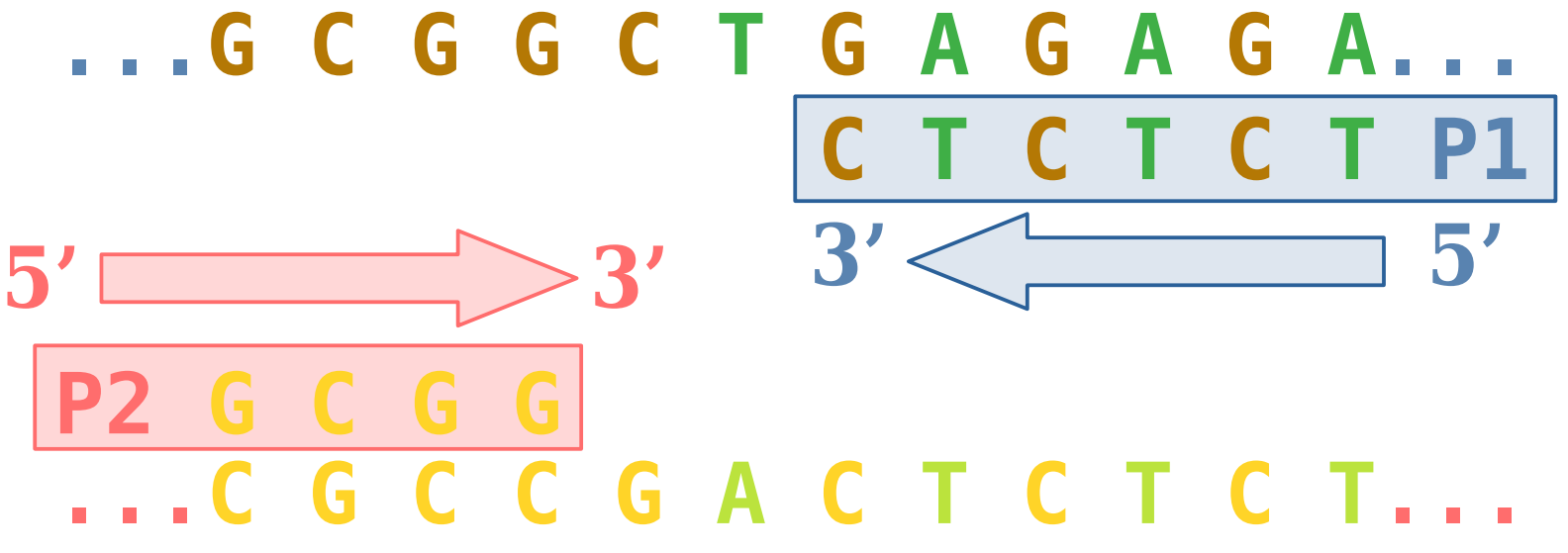
Separation of the 2 strands



PCR: Technique

Step 2: Hybridization of Primers

- **Temperature**: depends! (40 – 65º C)
- **Duration**: ~ 30 seconds;



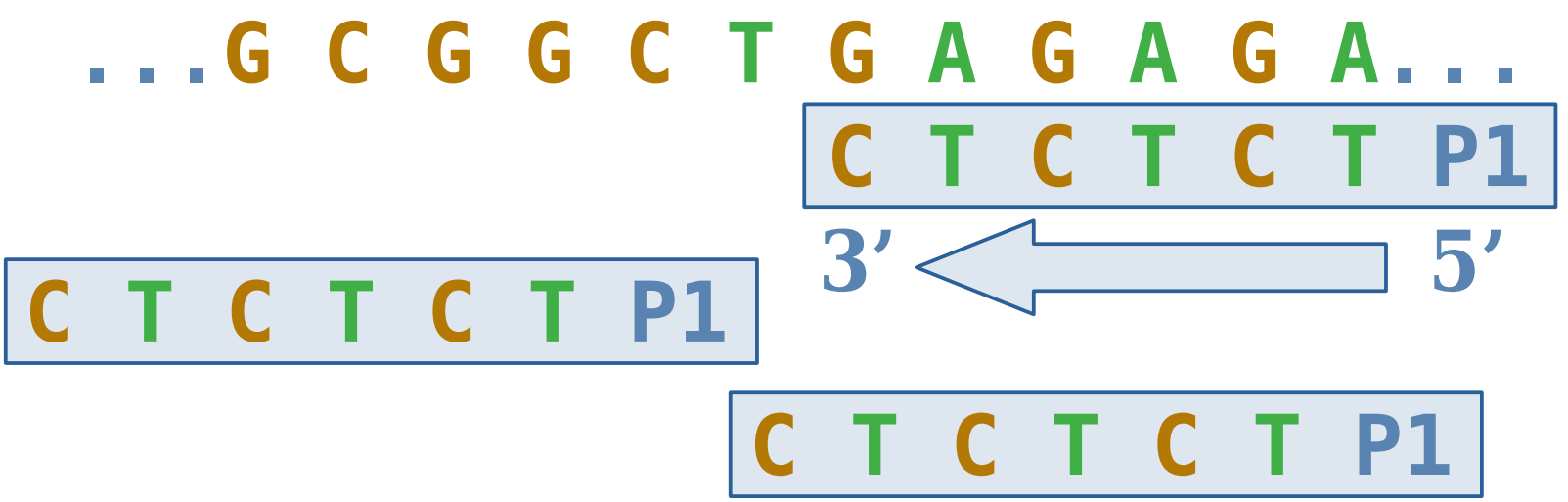
PCR: Technique

Step 2: Hybridization of Primers

- **Temperature**: depends! (40 – 65° C)
- **Duration**: ~ 30 seconds;

Primers:

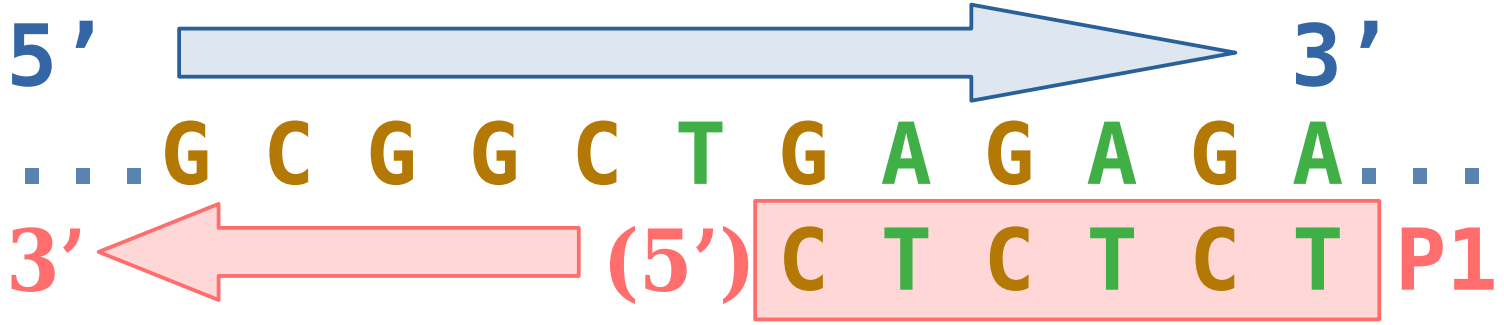
- **Excess** in reaction mixture;



PCR: Technique

Primers:

- **Length:** 15 – 25 Nts;
- **Temperature**: depends!
- **G+C**: ~ 50%;
- Synthesized chemically;
- Suppliers on web;
- **Tm** similar for both;



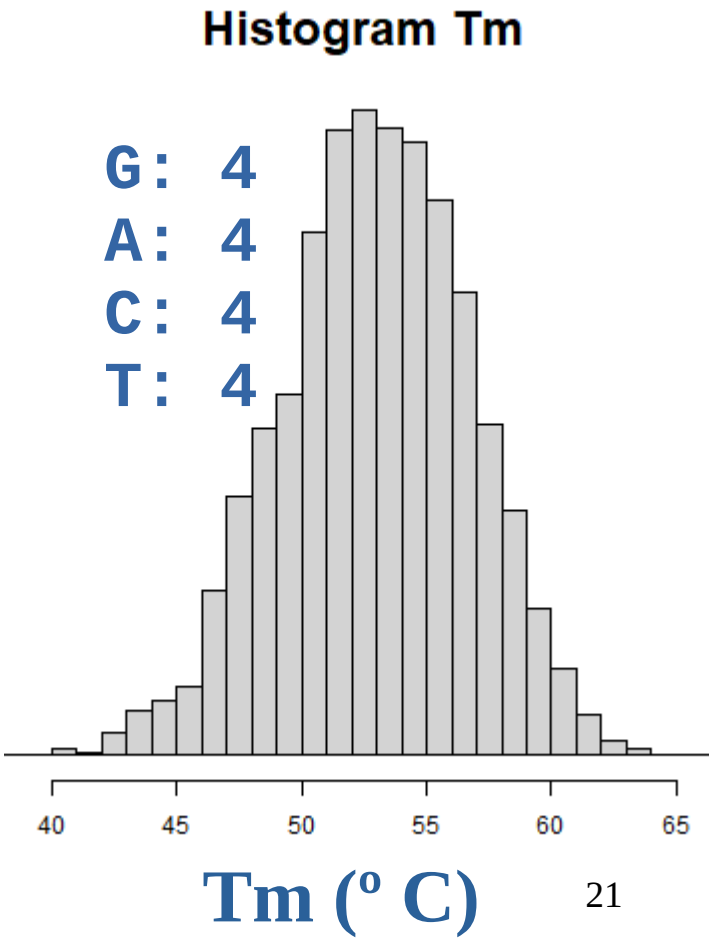
PCR: Primers

Tm of Primers: depends

- **Tm** = melting temperature;
- Sequence (neighboring Nt);
- **Ionic concentration!**

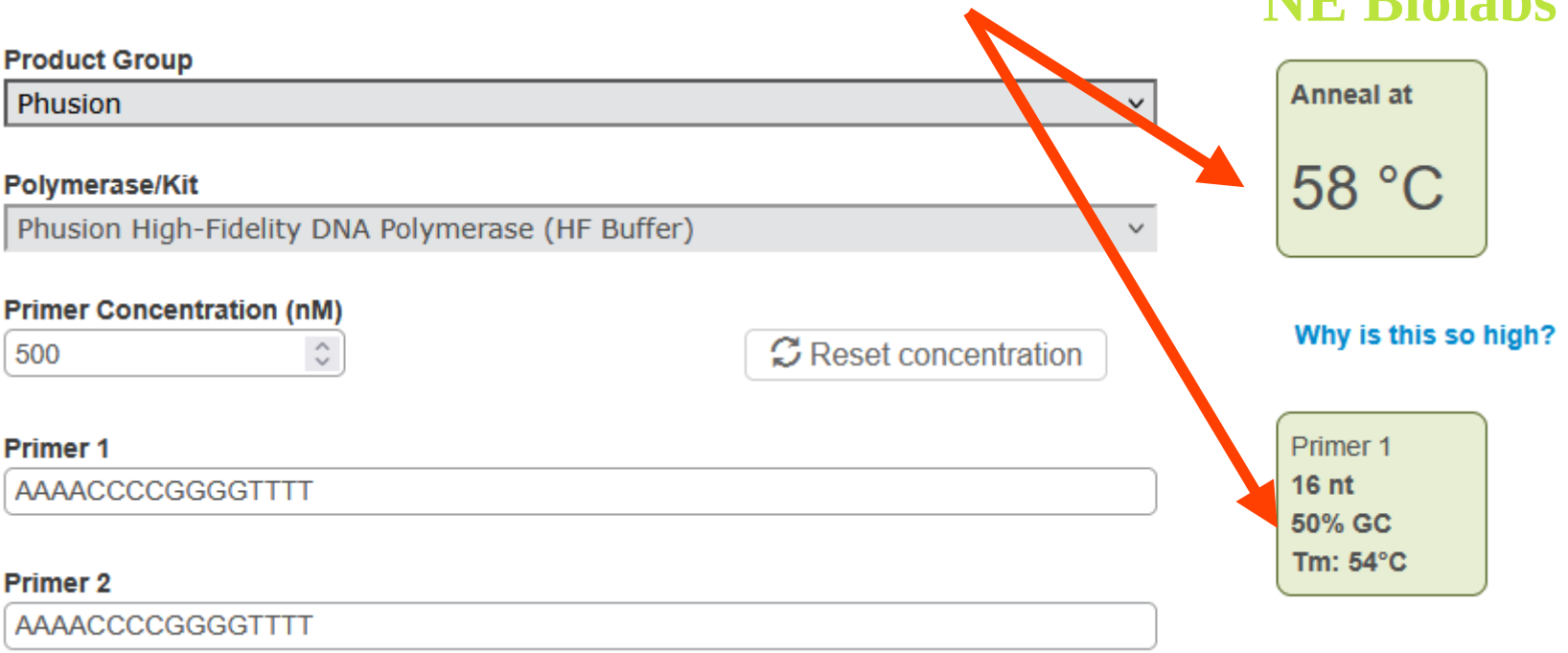
- NE Biolabs: Q5 60º C;
- NE Biolabs: OneTaq 46º C;
- Detailed formula: 61.5º C;

AAAACCCCGGGGTTTT



PCR: Primers

Temperature of Primers: depends



AAAACCCCGGGGTTTT

https://tmcalculator.neb.com/#!/main

PCR: Primer Problems

Problems / Troubleshooting:

- **NO** amplified fragment:
  - **Hybridization T:** too high;
  - Other causes: **NO** genetic material, Pol-inhibitors?
- **Nonspecific** amplification;
  - Multiple fragments, blurred bands;
  - **Hybridization T:** too low;
- **False Positives**: DNA Contamination?

Pol-inhibitors: e.g. phenol (used in extractions);

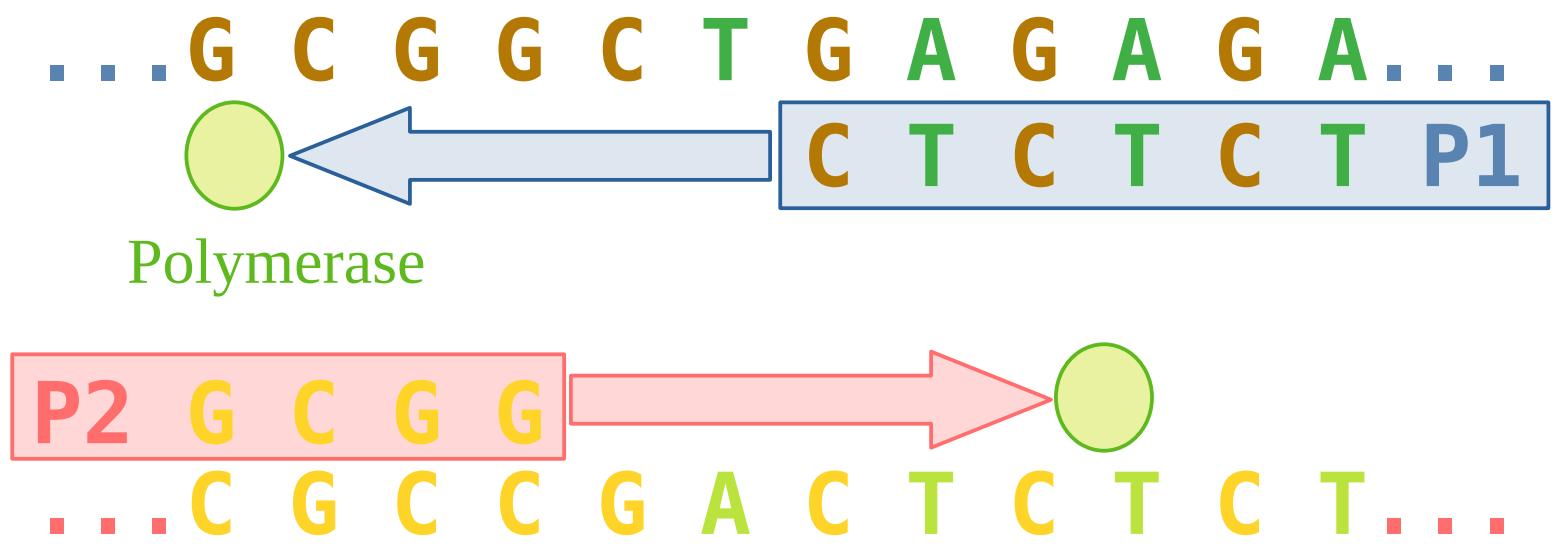
PCR: Technique

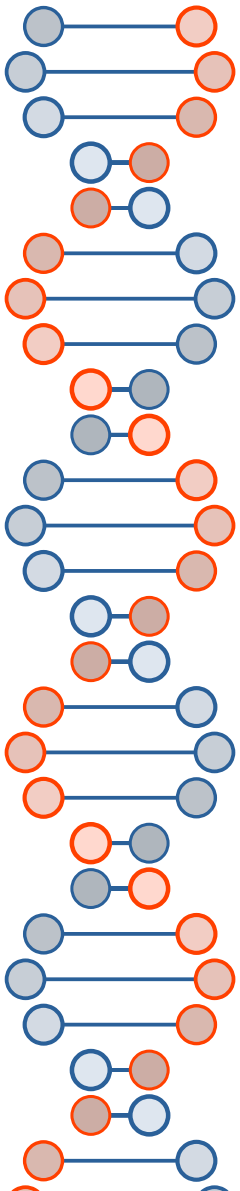
Step 3: Elongation

- **Temperature**: 72º C;
- **Duration**: ~ 1 minute / kb;

Polymerase:

- **Taq**: most frequent;
- **Pfu**, others;





PCR: Polymerase Enzyme

Taq: most frequent

- **Thermophilus aquaticus**: in “hot springs” (thermal water);
- **Temperature**: 72º C (optimal: 75 – 80º C);
- **Rate**: ~ 100 bp / s (at 72º C);
  - only 60 bp / s (at 70º C);
- **Stability**: $T_{1/2}$ = 40 min (at 95º C);
  - Drops rapidly at higher temperatures;

NO proof-reading!

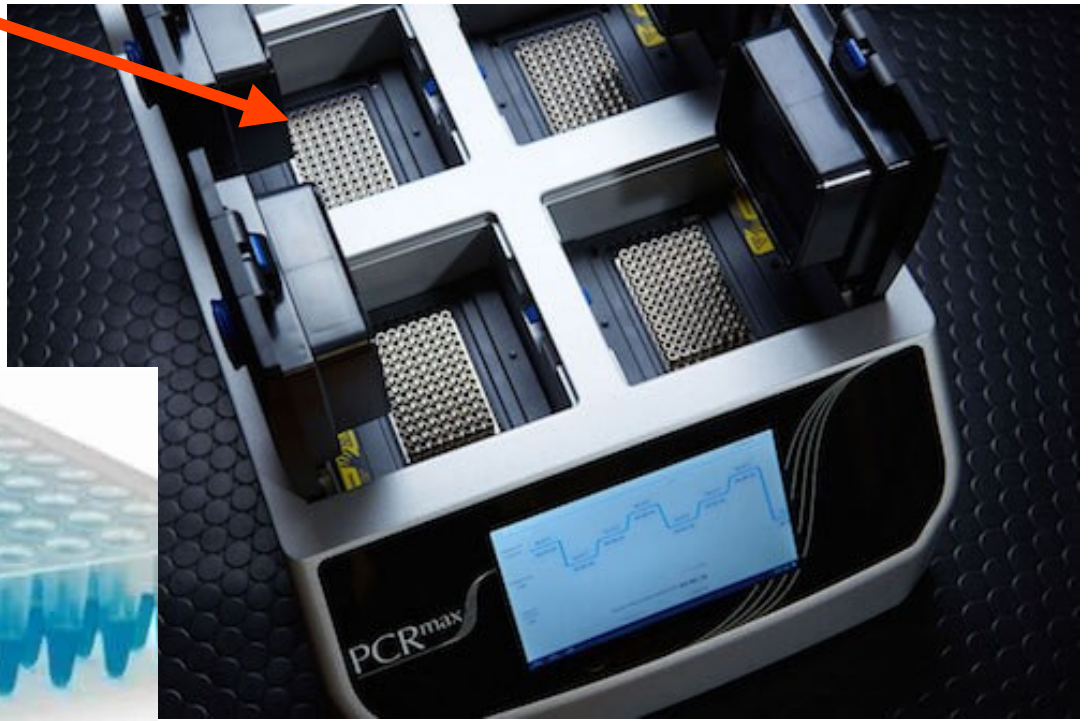
For proof-reading

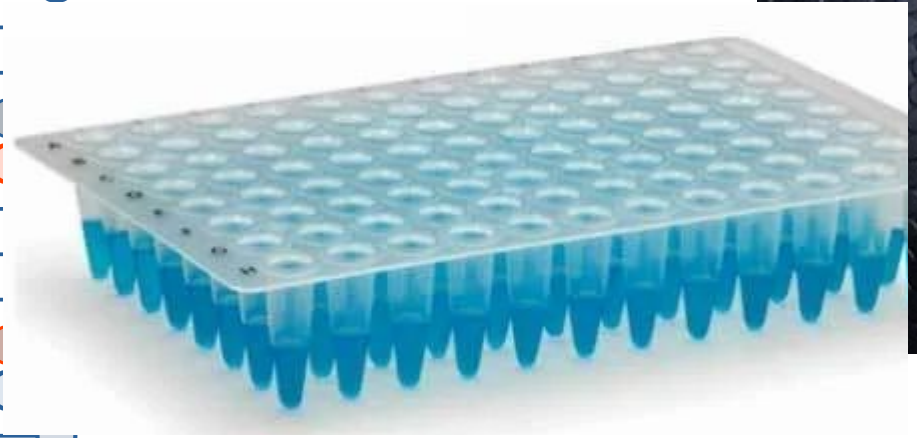
Other polymerases: **Pfu** (Pyrococcus furiosus);

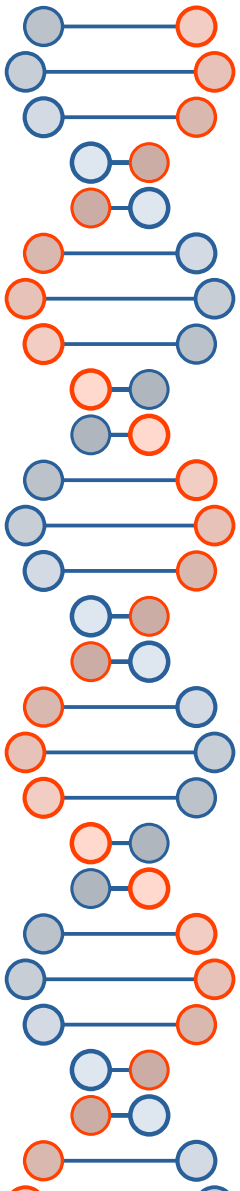
PCR: Thermocycler

Metallic plate:

- **Rapid changes of temperature!**







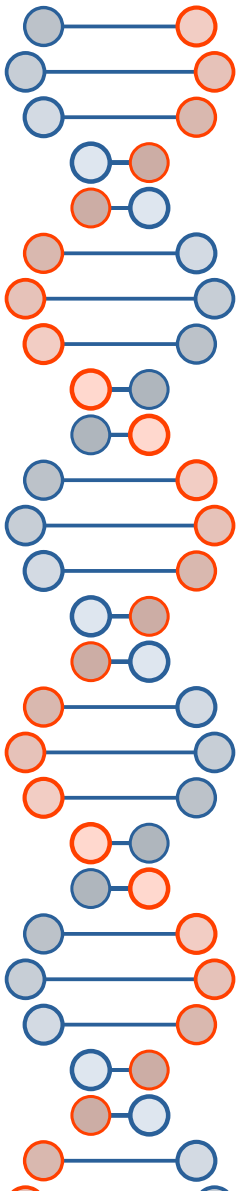
PCR: Time to Run

PCR Cycle:

- **Step 1**: **Denaturation** of DNA; 92 – 95º C; 30 s
- **Step 2**: **Hybridization** of primers; 40 – 65º C; 30 s
- **Step 3**: **Extension** of primers; 68 – 72º C; 1 min.
- Repeat: 30 cycles;

Total Time: ~ 2 hours

- 30 * 2 min = 60 min;
- additional + 5 min (pre-/post-) + pre-heating/cooling;



PCR: DNA Amplicon

Initially: Exponential amplification

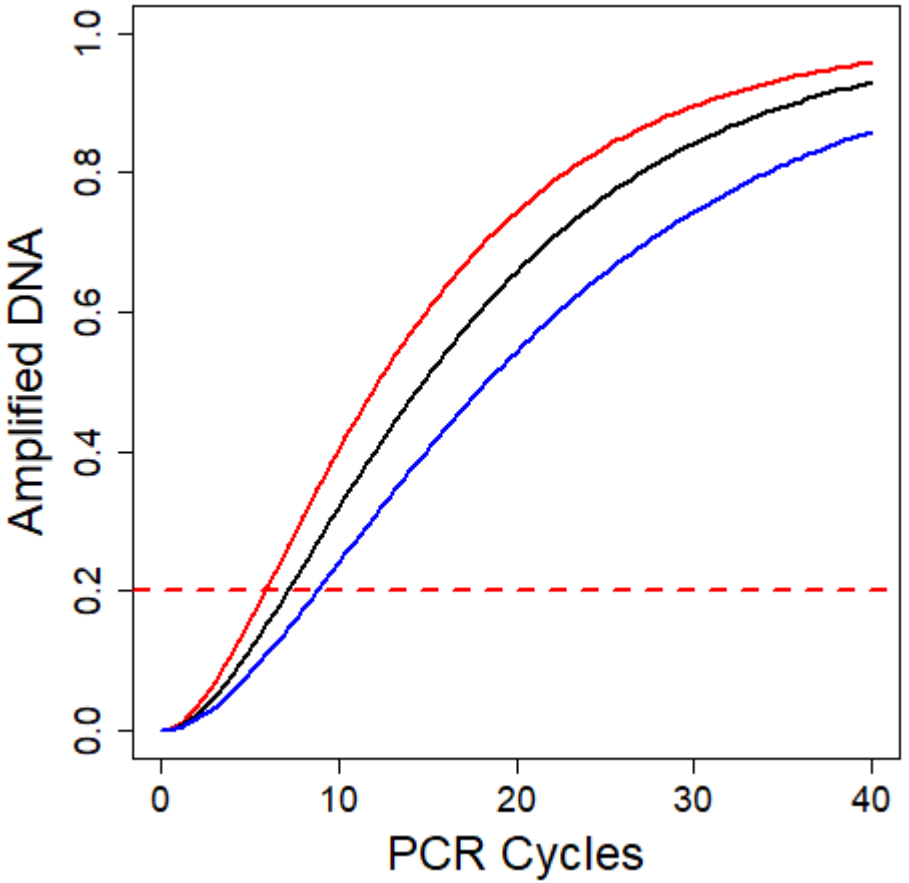
- Each cycle: **doubling** of product;
- Repeat: 30 cycles => 2^30 ≈ **1 billion**;

In time:

- **Leveling off:** reagents are consumed;
- **Plateau:** depletion of reagents (dNTP, primers);
- **Taq polymerase:** activity drops after 30 cycles;

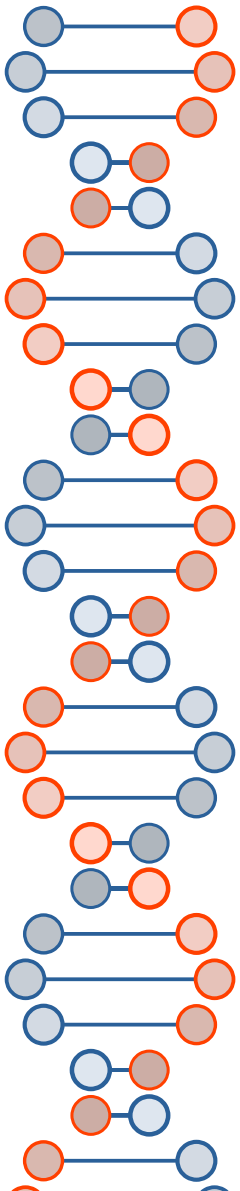
Product = also called amplicon;

PCR: DNA Amplicon



In time:

- **Leveling off:** reagents get consumed;
- **Plateau:** depletion of reagents (dNTP, primers);
- **Taq polimerase:** activity drops after 30 cycles;



PCR: Special Variants

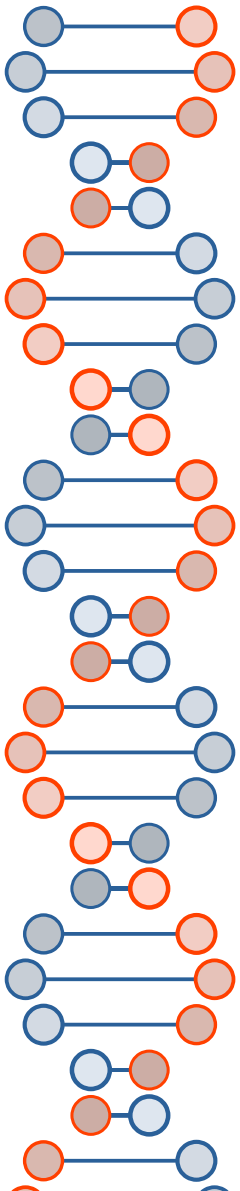
PCR: Special Variants

Reverse Transcription PCR (rtPCR):

- **Step 0: Revers transcription;**
- **RNA** => **cDNA** => **ds cDNA** => PCR on DNA;

Applications:

- **Viral RNA:** diagnosis of viral infections (ex. Covid, HCV);
- **Transcriptomics:** active genes => mRNA;



PCR: Special Variants

Quantitative PCR (qPCR):

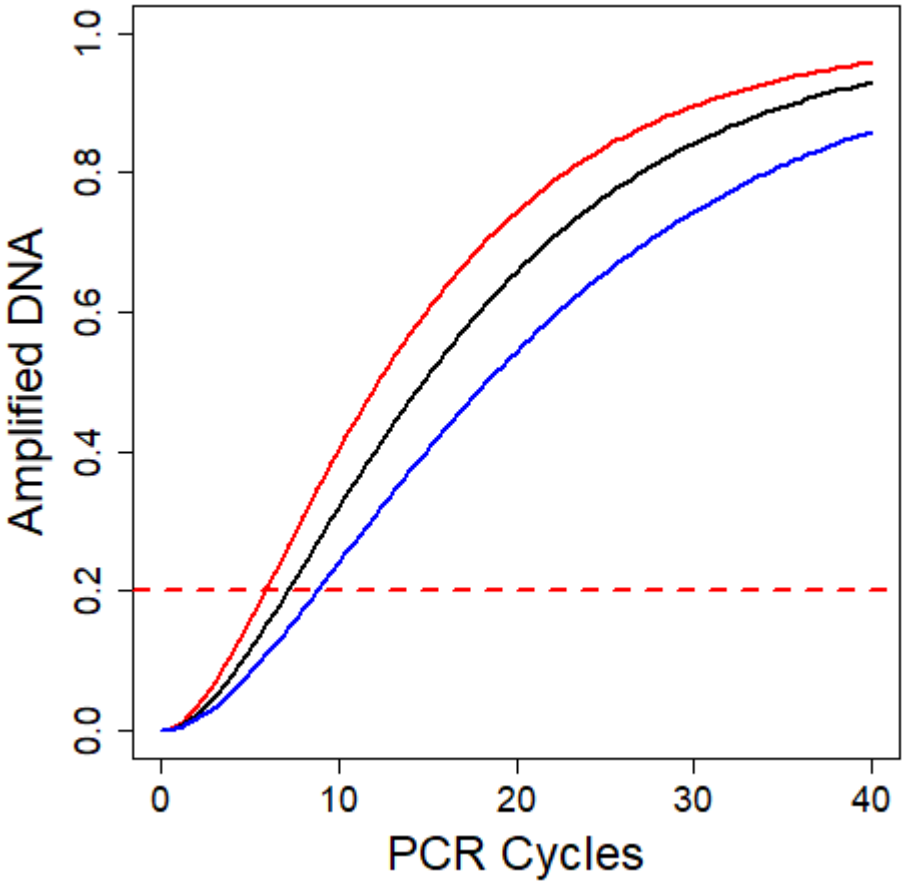
- qPCR = Real Time PCR;
- **Starting Material**: frequent **mRNA**

  (rt-qPCR in **Transcriptomics**);

Principles:

- **Visualization** of amplicon in real time;
- **Fluorophores**: various types;

qPCR: Quantitative PCR



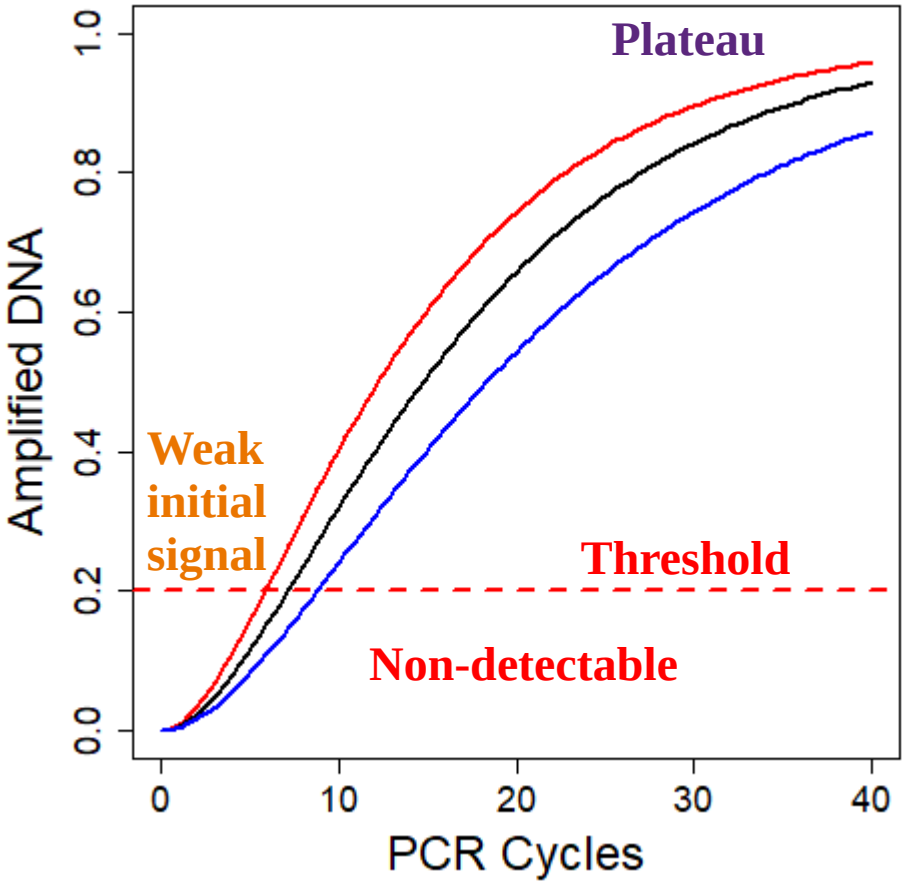
Quantifying DNA:

- **When to measure?**

In time:

- **Leveling off:** reagents consumed;
- **Plateau:** reagents depleted (dNTP, primers);

qPCR: Quantitative PCR



Quantifying DNA:

- **When to measure?**

  Continuously: **Real-time**
- Detectable signal: when above **threshold**;
- **Cycle threshold** (CT): number of cycles to cross **threshold**;

qPCR: Quantitative PCR

Principles:

- **Visualization** of amplicon in real time;
- **Cycle threshold (CT):**

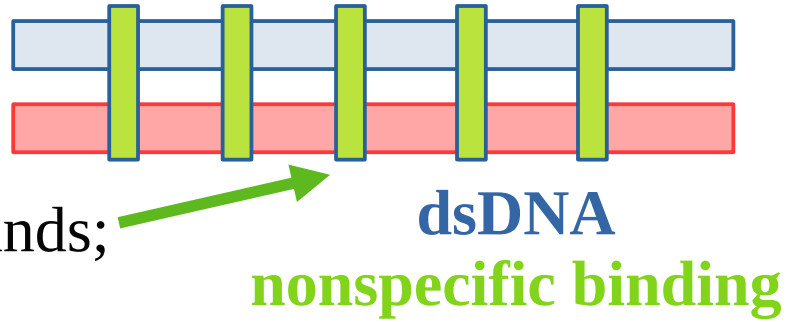
  number of cycles to cross **threshold**;
- **Fluorophores**: various types;

Quantitative PCR

Fluorophores

SYBR Green:

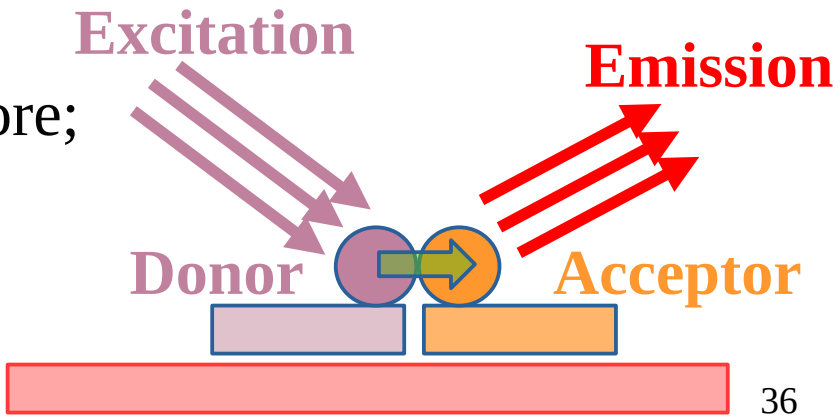
- **Intercalates** between DNA strands;



DNA Probes, FRET:

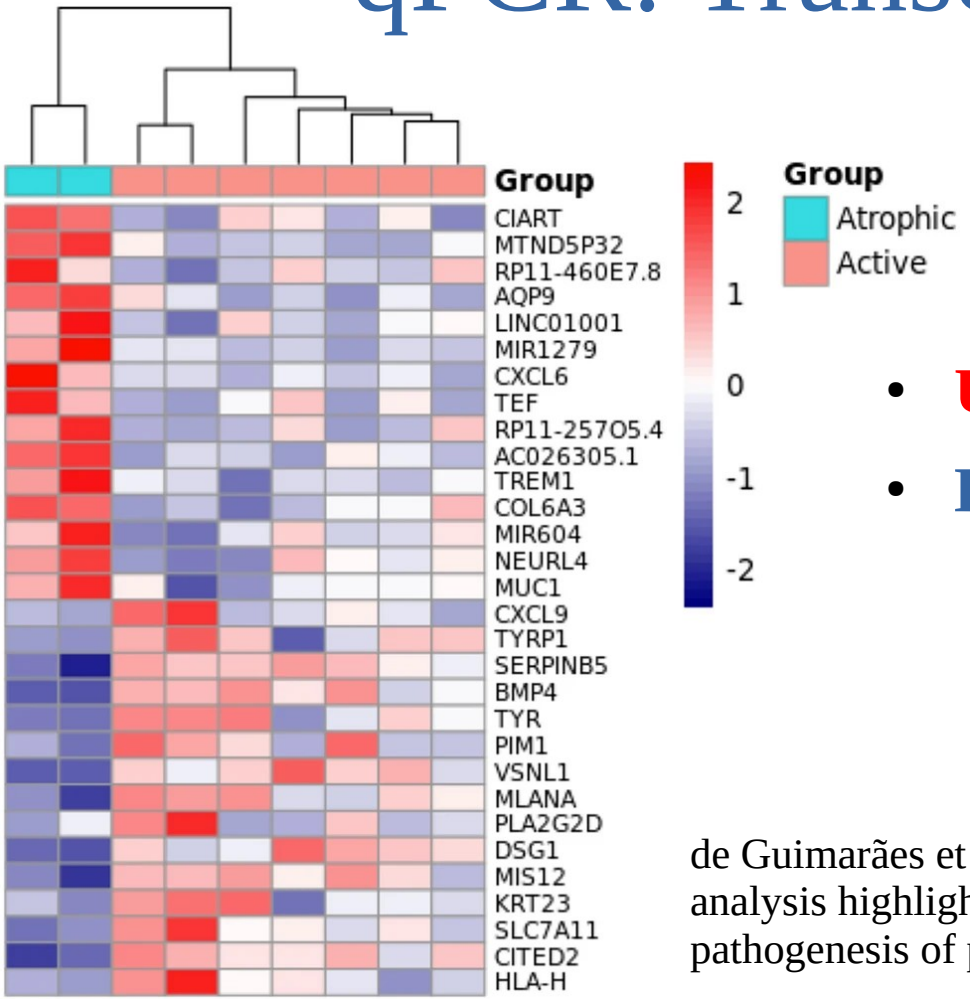
- Primers marked w. fluorophore;
- **Spatial Proximity** =>

  Energy Transfer (ET);



Navarro E et al. Real-time PCR detection chemistry. Clin Chim Acta. 2015

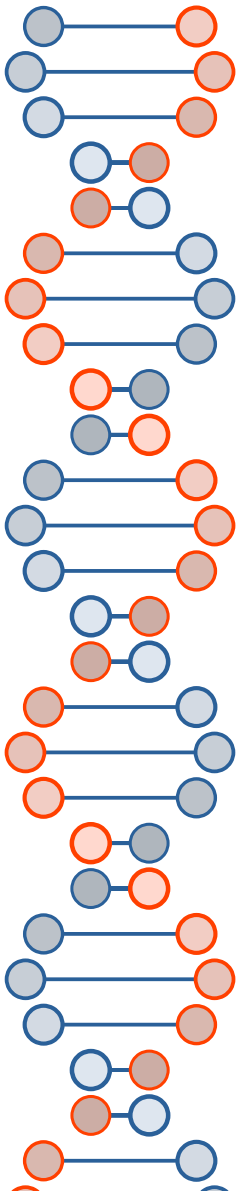
qPCR: Transcriptomics



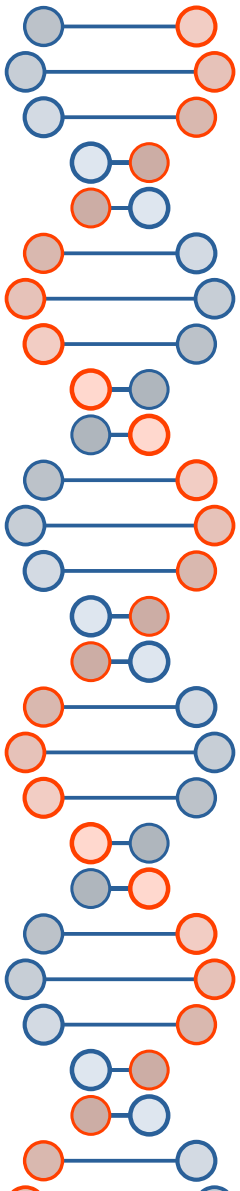
Differential Gene Expression:

- **Upregulated genes;**
- **Downregulated genes;**

de Guimarães et al. Transcriptomics and network analysis highlight potential pathways in the pathogenesis of pterygium. (2022)



PCR: Applications



PCR: Applications

Medicine:

- **Diagnostics**: **genetic** or **infectious** diseases;
- **Monitoring**: quantifying **HIV** virus, **genotyping**;

Molecular Biology / Biotechnologies:

- Multiple applications: cloning of genes, etc;

Forensics:

- **Identification** of persons, **paternity** test, etc;

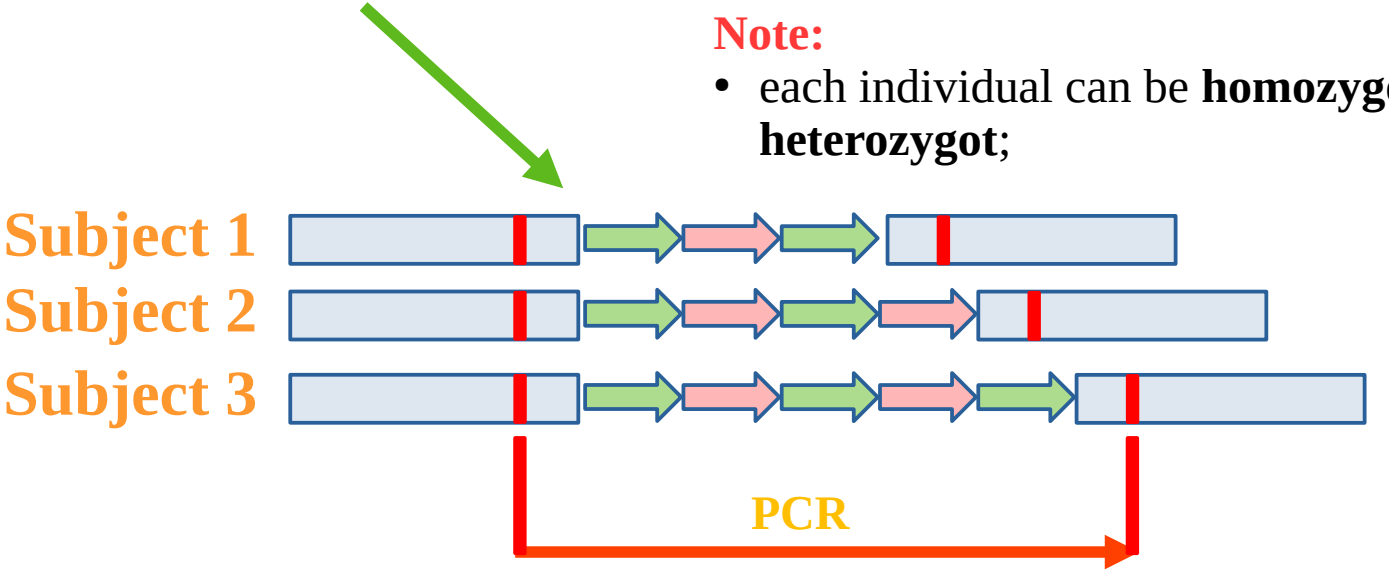
Forensics: Short Tandem Repeats

Short Tandem Repeats (STR):

- Short DNA sequence: 2 – 6 bp, ex (GATA) or (TCTA);
- Variable **repetitions** in population: 3 – 20 times;

Note:

- each individual can be **homozygot** or **heterozygot**;

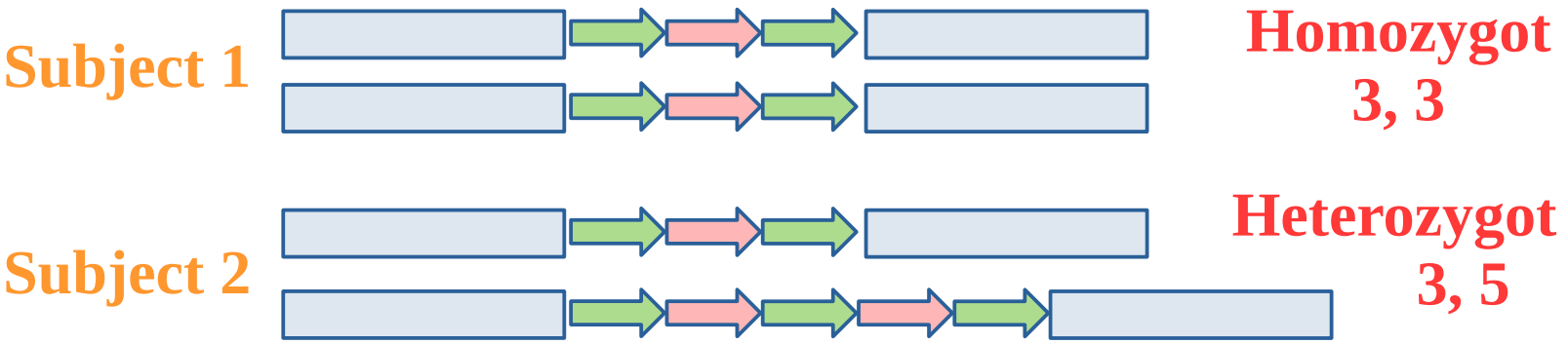


...

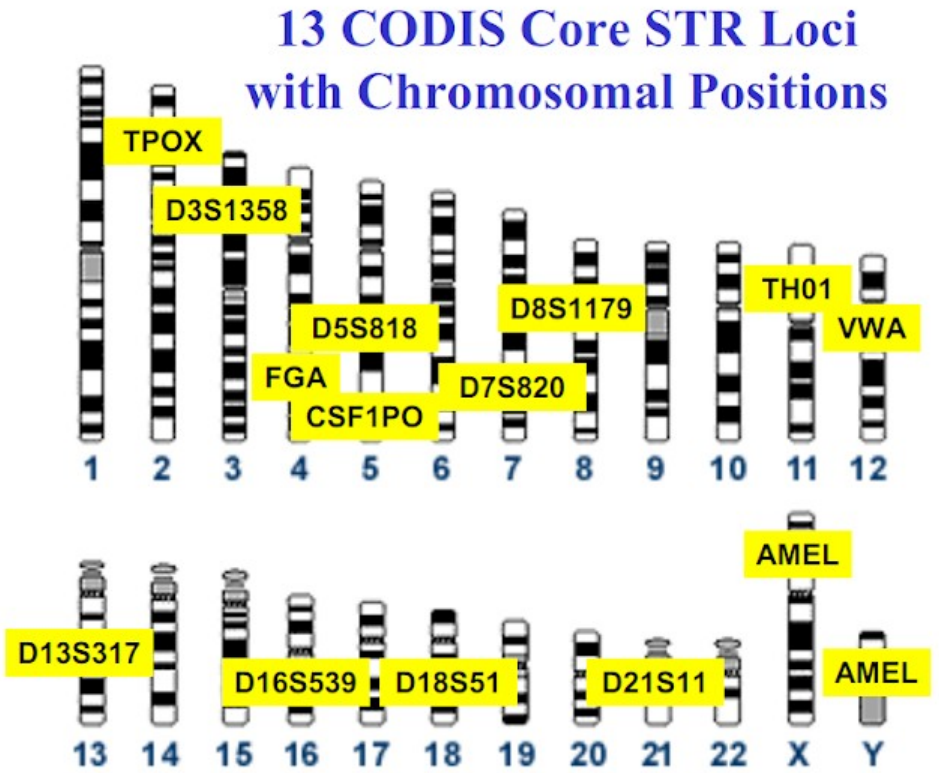
Forensics: Short Tandem Repeats

Short Tandem Repeats (STR):

- Short DNA sequence: 2 – 6 bp, ex (GATA) or (TCTA);
- Variable **repetitions** in population: 3 – 20 times;



Forensics: Short Tandem Repeats



Loci:

- FBI: 13 loci;

  (meanwhile more)
- Special 2: AMEL for Chr X, Y;
- Many combinations homozygote & heterozygote;

Forensics: Short Tandem Repeats

Supplementary Material (Example)

Table 2

Summary of STR locus characteristics including repeat motif, allelic size ranges, number of observed, potential and missing alleles and expected heterozygosity (He)

STR Loci in Plants (CS = Canabis sativa)

STR locus	ANUCS201	ANUCS301	ANUCS302	ANUCS303	ANUCS202
STR motif	$(GA)_{26}$	$(TTA)_{15}$	$(CAA)_{11},\cdots,(CAA)_4$	$(TGG)_7$	$(GA)_{20}$
Repeat unit	2	3	3	3	2
Size of sequenced STR	187	234	147	147	161
Min allele size (bp)	155	209	140	142	145
Max allele size (bp)	227	261	173	157	185
Inferred min STR	$(GA)_{10}$	$(TTA)_7$	$(CAA)_9$	$(TGG)_5$	$(GA)_{12}$
Inferred max STR	$(GA)_{46}$	$(TTA)_{24}$	$(CAA)_{20}$	$(TGG)_{10}$	$(GA)_{32}$
No. of observed alleles	29	16	10	6	18
No. of potential alleles	37	18	12	6	21
No. of missing alleles	8	2	2	0	3
He	0.928	0.902	0.758	0.659	0.902

All estimates are based on the total data set of 93 samples. Number of potential alleles were inferred by counting the number of repeat units between the smallest and largest observed alleles. The number of missing alleles were inferred by subtracting the number of observed alleles from the total number of potential alleles within the observed allele range.

Gilmore S et al. (2003) Short tandem repeat (STR) DNA markers are hypervariable and informative in Cannabis sativa: implications for forensic investigations.

PCR: Other Applications

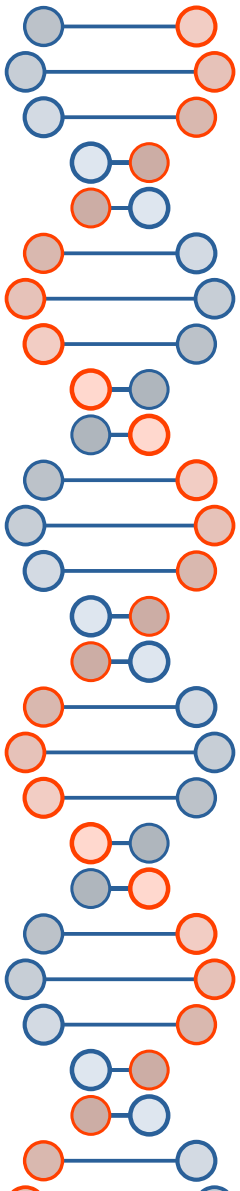
Metagenomics:

- Analysis of **bacterial communities;**
- E.g.: in **dental plaques** or the **gut microbiota**;

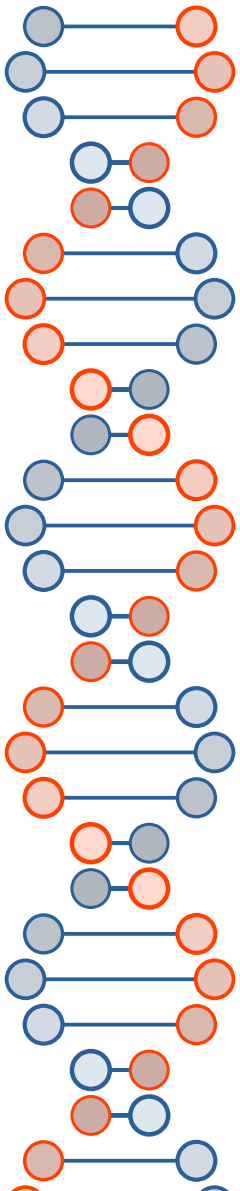
Genome Sequencing:

- 2 Methods;
- **Shotgun sequencing** vs **PCR-based sequencing**;

Pang L et al. Metagenomic Analysis of **Dental Plaque** on Pit and Fissure Sites With and Without Caries Among Adolescents. Front Cell Infect Microbiol. 2021



Supplementary Material



Supplementary Material

Variability of Tm: Primers 4 x G, 4 x C, 4 x A, 4 x T

- **Tm**: depends on **explicit sequence**;(*)
- **AAAACCCCGGGGTTTT**: 61.5º C
- TGGCATGCTTCCAGAA: 60.0º C
- CCAGGTTCGGCATATA: 55.0º C
- AGTCGACATTCAGCTG: 52.0º C
- CCAGGAGTCTACTAGT: 42.0º C
- GTAGTCCGTACAGTAC: 40.1º C

Length:

- Usually: 16 –25 nucleotides;

* theoretical Tm computed using extended formula;

Supplementary Material

Hot springs:

Yellowstone

- **Thermophilus aquaticus**

